



SECOND SEMESTER 2024-25
COURSE HANDOUT

Date: 06.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : ECON F413
Course Title : FINANCIAL ENGINEERING
Instructor-in-Charge : Prof. Aditya Sharma
Instructor(s) : Prof. Byomakesh Debata, Prof. Aditya Sharma
Tutorial/Practical Instructors: NA

1. Course Description:

The Financial Engineering Concentration encompasses the design, analysis, and construction of financial contracts to meet the needs of enterprises. It involves developing and creatively applying financial theory and financial instruments to structure solutions to complex financial problems and exploit financial opportunities.

2. Scope and Objective of the Course:

The objective of the course is to provide a preliminary understanding of financial engineering tools and techniques in the present market environment, such as applications of futures and options for hedging, asset price dynamics, and the use of simulation in decision-making. Financial Engineering is a multidisciplinary field involving financial theory, the methods of engineering, the tools of mathematics, and the practice of programming. The course on financial engineering would enable students to respond to the professional requirements particularly related to investment banking, financial management, treasury operations, asset management, risk management, consulting and capital markets, etc.

3. Text Books:

T1. Salih N. Neftci, Principles of Financial Engineering, Academic Press Publishers, 2009

4. Reference Books:

R1. David A. Dubrofsky and Thomas W. Miller, Jr., Derivatives Valuation and Risk Management, Oxford University Press.

R2. David G. Luenberger, Investment Science, Oxford University Press

R3. Futures, Options and Other Derivatives by John C Hull, 8th Edition, Prentice Hall



5. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
1	1-2	Chap 1+ Class notes	Understanding the concept of financial engineering, its tools and techniques etc.
3	3-13	Class notes+ Chap3 & 4 of T1 + R3	To know about the fixed income securities. Understanding the concepts of term structures and fixed income derivatives. Determination of interest rate Forwards and Futures prices, Cash flow engineering. Special topic: Interest rate derivative modeling
4	14-18	Class Notes + Chap 9 of T1	Pricing Bonds, Learning convexity of bond and yield volatility.
5	19-29	Chap 5 & 13 of T1+R1+ R3 + class notes	Understanding the concepts of swaps and their different types and uses. Swaps engineering.
6	30-39	Chap 8 & 10 of T1, + R3 plus class notes (Pre-read: Wiener process, BSM model)	Understanding the concepts of Options Pricing, Option mechanics, options engineering and its application.
7	40-42	Chapter 11&12 + Class notes	Understanding VaR, Learning Tools of financial engineering in pricing and application of fundamental theorems Special topic: Monte-Carlo simulation and its application in Financial Engineering



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6. Evaluation Scheme:

Component	Weightage (%)	Date and Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	30	As per timetable	CB
Comprehensive Examination	40	As per timetable	OB/CB
Assignments / Labs / Quizzes	30	TBD	Continuous

7. Chamber Consultation Hour: 4:00-5:00 PM - Friday

8. Notices: All notices will be displayed on the Department of Economics & Finance Notice Board

9. Make-up Policy: Make-up will be given only on genuine cases and on prior approval.

10. Note (if any):

Instructor-in-charge